Health Economics

Three Chapters in Sequence

Chapter 1: Health Policy

Chapter 2: The SIR Model of Infectious Disease

Chapter 3: Pharmaceutical Innovation

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Course Outline: Three Chapters

Chapter 1 — Health Policy

Health-system scarcity; Beveridge and Bismarck systems; insurance; adverse selection and moral hazard; population aging; fiscal sustainability and waiting times.

Chapter 2 — Economic Epidemiology and the SIR Model

Susceptible–infectious–recovered dynamics; transmission, reproduction numbers, epidemic peaks, disease prevention, externalities, and hospital congestion.

Chapter 3 — Pharmaceutical Innovation

Drug life cycle, development uncertainty and costs, patents, induced innovation, regulation, technology, pricing, and health expenditure growth.

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Health Policy

Beveridge and Bismarck Systems, Insurance, Population Aging

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Health Policy: Fundamental Economic Questions

Health-care systems face the same economic problem as other activities: human needs are large, but resources are scarce.

1. Who finances treatment and which risks are pooled?
2. Who provides health services and who regulates providers?
3. How much care should be delivered at the margin?
4. How is treatment distributed when medical capacity is scarce?
5. How can the system remain financially sustainable as populations age?

Core distinction

Financing determines *who pays*; provision determines *who treats*; allocation determines *who receives treatment and when*.

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Two Major Health-Care Systems

	Beveridge-type system	**Bismarck-type system**
Financing	General taxation	Social health-insurance contributions
Institutions	Central role for government	Sickness or social-insurance funds
Provision	Often substantial public provision	Often substantial private provision
Typical allocation issue	Budget constraints, waiting and priority	Risk pooling, insurance incentives and provider payments

Institutional qualification

Actual countries are hybrids. Financing and provision need not be either wholly public or wholly private.

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The Beveridge Model: Definition and Features

A Beveridge-type system relies primarily on **general taxation** to finance broad access to health care.

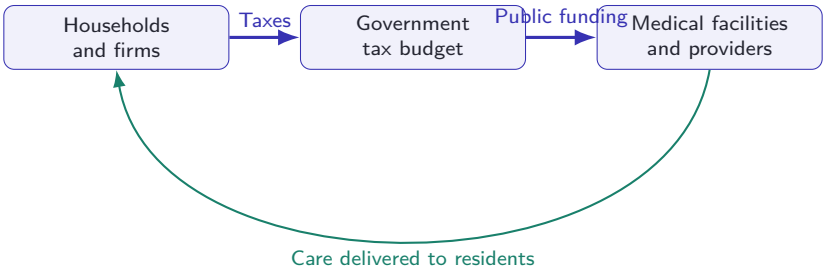
- Government plays a central role in financing and often in service delivery.
- Patients may face low or zero fees at the point of service.
- Health care remains resource-intensive even when monetary patient prices are zero.
- Common illustrations include the United Kingdom's NHS; other systems incorporate Beveridge features.

Economic tension

Low patient prices facilitate access, but providers remain constrained by staff, beds, operating budgets, equipment and time.

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How Beveridge Financing Works



Point-of-use and resource prices are distinct

Even when the patient's bill is zero, the economy gives up labor, equipment and alternative goods to supply treatment.

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Consumer Choice: Ordinary Goods and Health Care

Suppose the consumer chooses ordinary consumption c and health-care use h to maximize

$$U(c, h) = u(c) + v(h), \qquad c + ph = y.$$

Here p is the out-of-pocket price per unit and y is income. Assume $u'(c) > 0$, $v'(h) > 0$, and diminishing marginal benefit $v''(h) < 0$ over the relevant interior range. Substitute $c = y - ph$ into utility:

$$U(h) = u(y - ph) + v(h).$$

Differentiate by h:

$$\frac{dU}{dh} = -p\, u'(y - ph) + v'(h).$$

For an interior optimum,

$$\boxed{v'(h^*) = p\, u'(c^*).}$$

The consumer equates marginal health benefit with the utility value of the sacrificed consumption.

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Consumer Demand When the Patient Price Is Zero

Starting with the optimality condition,

$$v'(h^*) = p\,u'(c^*),$$

set $p = 0$. Then an *interior satiation point*, if one exists, satisfies

$$v'(h^*) = 0.$$

If marginal utility remains strictly positive for every finite h, a zero price instead pushes desired care toward the consumption or capacity bound; there is no finite interior solution.

Economic interpretation

Health-care use can rise sharply as out-of-pocket prices fall. Zero patient payment is not the same as zero social resource cost.

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A Zero Patient Price Can Create Excess Demand

Let $H_D(p)$ be aggregate demand, and let $\bar{H}$ be the maximum feasible number of services. At a zero point-of-use price,

$$H_D(0) > \bar{H} \quad \text{may hold.}$$

Subtract capacity from demand:

$$\boxed{ED = H_D(0) - \bar{H} > 0.}$$

This is excess demand, not extra physical capacity. An allocation mechanism is needed to determine whose care is provided first.

Practical mechanisms

Waiting time, medical priority, referral requirements, provider budgets, admission criteria and treatment guidelines.

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Rationing Without Money Prices

- **Queues:** an appointment or procedure requires waiting.
- **Clinical prioritization:** urgent cases are treated before less urgent ones.
- **Gatekeeping:** referrals determine access to specialist care.
- **Service restrictions:** guidelines and budgets limit certain treatments.
- **Capacity investment:** additional personnel, beds and equipment relax scarcity over time.

Central lesson

$$\text{No monetary price} \not\Rightarrow \text{No scarcity.}$$

Public provision changes how the resource constraint is managed; it does not repeal that constraint.

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Waiting Time as an Implicit Price

Let w be time spent waiting, and let $C(w)$ be the nonmonetary cost of waiting. Suppose

$$C'(w) > 0.$$

Let health-care demand depend negatively on that cost:

$$H_D = H_D(C(w)), \qquad \frac{\partial H_D}{\partial C} < 0.$$

By the chain rule,

$$\frac{dH_D}{dw} = \frac{\partial H_D}{\partial C}\frac{dC}{dw}.$$

The first factor is negative and the second positive. Therefore,

$$\boxed{\frac{dH_D}{dw} < 0.}$$

Longer waiting increases the effective cost and, other things equal, discourages utilization.

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How an Equilibrium Waiting Time Emerges

Suppose the health-care system has fixed capacity $\bar{H}$. An equilibrium wait w^* is defined by

$$\boxed{H_D(C(w^*)) = \bar{H}.}$$

1. If demand exceeds capacity, waiting and congestion tend to rise.
2. As waiting cost increases, some marginal demand is postponed or forgone.
3. Under continuity and suitable monotonicity, the adjustment can settle at demand equal to available capacity.

Comparison

An ordinary market can use a changing monetary price; a capacity-constrained public service can use waiting time and priority rules as part of its allocation system.

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Efficient Health Care: Marginal Benefits and Costs

Suppose the marginal social benefit and marginal resource cost of health care are

$$MB(h) = 10 - h, \qquad MC(h) = 4.$$

Efficiency requires $MB(h^*) = MC(h^*)$:

$$10 - h^* = 4,$$

$$10 - 4 = h^*,$$

$$\boxed{h^* = 6.}$$

At the efficient quantity, an extra unit's marginal benefit equals the value of resources needed to produce it.

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Deriving Social Surplus Explicitly

From $MB(h) = 10 - h$, total benefit is the integral of marginal benefit:

$$\begin{aligned} TB(h) &= \int_0^h (10 - x)\, dx \\ &= \left[10x - \frac{x^2}{2}\right]_0^h \\ &= 10h - \frac{h^2}{2}. \end{aligned}$$

Since $MC(h) = 4$, total resource cost is $TC(h) = \int_0^h 4\, dx = 4h$. Hence,

$$\begin{aligned} W(h) &= TB(h) - TC(h) \\ &= \left(10h - \frac{h^2}{2}\right) - 4h \\ &= 6h - \frac{h^2}{2}. \end{aligned}$$

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Maximizing Social Surplus

Take the derivative of $W(h) = 6h - h^2/2$:

$$W'(h) = 6 - h.$$

The first-order condition is $W'(h^*) = 0$, which yields

$$6 - h^* = 0 \implies h^* = 6.$$

The second derivative is

$$W''(h) = -1 < 0,$$

so the critical point is a strict maximum.

Compare with zero patient price

If private demand stops where $MB(h) = 0$, then $h_D(0) = 10$ and $h_D(0) - h^* = 4$. The illustrative difference is potential overutilization relative to efficient resource use.

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The Bismarck Model: Definition and Features

A Bismarck-type system is organized primarily around **compulsory social health insurance**.

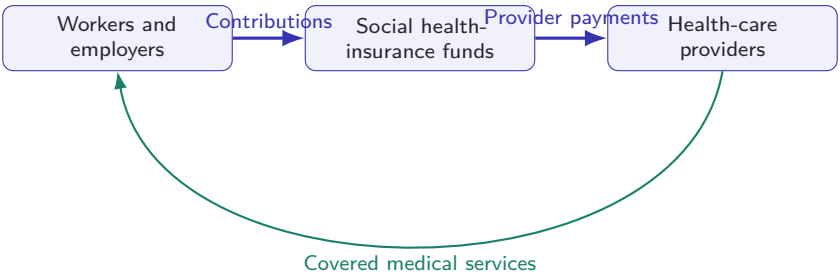
- Contributions are conventionally related to earnings and often shared by employers and employees.
- Funds collect contributions, pool risks and pay medical providers.
- Providers may be independent or privately organized.
- Government regulates participation, benefit rules and payment arrangements.
- Germany is the conventional reference case; real national arrangements differ.

Central economic issues

Risk pooling, insurance incentives, contribution rates, and the financial burden of medical spending.

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How Bismarck Financing Works



Distinction

Patients receive care when ill, while many participants contribute to a common pool when healthy.

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Why Individuals Demand Insurance

Suppose income before medical expenses is Y. Illness occurs with probability q, causing a loss L. Without insurance, consumption in the two states is

$$C_H = Y, \qquad C_S = Y - L.$$

The probability of remaining healthy is $1 - q$. Assume risk aversion:

$$u'(C) > 0, \qquad u''(C) < 0.$$

Expected utility without insurance is

$$\boxed{EU_0 = (1 - q)u(Y) + q\,u(Y - L).}$$

Even if expected income loss is known, actual consumption remains uncertain.

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Expected Utility with Insurance

Let J denote an insurance indemnity paid only when sick, and let P be the premium.

$$C_H = Y - P,$$

$$C_S = Y - P - L + J.$$

Expected utility with insurance is

$$\boxed{EU_J = (1-q)u(Y-P) + q\,u(Y-P-L+J).}$$

Insurance is privately attractive if $EU_J > EU_0$.

Interpretation

Insurance transfers purchasing power from the healthy state to the sick state. Its value is especially high for a person with concave utility.

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Actuarially Fair Insurance and Full Coverage

If the insurer exactly covers expected claims and ignores administrative costs, the actuarially fair premium is

$$P = qJ.$$

For full coverage $J = L$:

$$P = qL.$$

Healthy-state consumption becomes

$$C_H = Y - qL,$$

and sick-state consumption becomes

$$C_S = Y - qL - L + L = Y - qL.$$

Therefore,

$$\boxed{C_H = C_S = Y - qL.}$$

With state-independent strictly concave utility and fair pricing, fully smoothing consumption across the two states is optimal.

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Insurance, Risk Pooling, and Compulsory Participation

Suppose individual i has illness probability q_i and loss L_i. For N insured individuals, the expected medical expenditure is

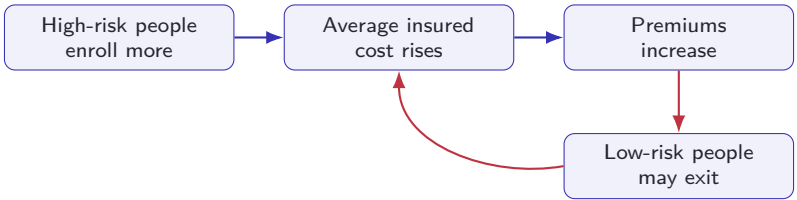
$$\mathbb{E}[M] = \sum_{i=1}^{N} q_i L_i.$$

- Insurance distributes uncertain losses across many contributors.
- Broad compulsory participation can prevent low-risk members from exiting.
- Greater risk pooling does not make total expected resource costs disappear.
- Contributions, benefits and redistribution are governed by policy choices.

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Adverse Selection in Health Insurance

Adverse selection arises when enrollees have more information about their risk than insurers, and coverage choices reflect that information.



Potential policy responses

Compulsory enrollment, subsidies, risk adjustment, common benefit design and public information standards.

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Moral Hazard: Insurance Changes the Patient Price

Let p be the real resource price of a treatment and α the fraction paid by the insured patient.

$$0 \leq \alpha \leq 1, \qquad p_c = \alpha p.$$

The effective monetary price facing the patient is p_c. Greater coverage means $\alpha \downarrow$. If treatment demand decreases with patient price,

$$\frac{\partial H_D}{\partial p_c} < 0,$$

then

$$\boxed{\alpha \downarrow \Longrightarrow \; p_c \downarrow \Longrightarrow \; H_D \uparrow .}$$

This utilization response is an example of **ex post moral hazard**; it need not imply waste, since some additional care can be valuable.

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Beveridge Versus Bismarck: Economic Comparison

Dimension	Beveridge	Bismarck
Revenue	General taxes	Social insurance contributions
Point-of-use price	Oftentimes low or zero	Determined by benefits and cost-sharing
Prominent constraints	Public budgets and medical capacity	Fund solvency and insured incentives
Common responses	Clinical priority, waiting, investment	Pooling, contribution adjustments, regulated payments

Shared objectives

Equity, financial protection, effective treatment, efficient resource allocation, and system sustainability.

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Population Aging: Definitions

Divide the population into younger (N_Y) and elderly (N_E) groups:

$$N = N_Y + N_E.$$

The elderly population share is

$$\boxed{s_E = \frac{N_E}{N}.}$$

Aging means s_E increases over time. Common demographic drivers include falling fertility, longer survival, and the aging of large cohorts. If N_W denotes the working-age contributing population, the old-age dependency ratio is

$$\boxed{D = \frac{N_E}{N_W}.}$$

The definitions of N_Y and N_W need not coincide; children can be part of N_Y without being contributors.

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Dependency Ratio and Elderly Share: Two-Group Case

In the illustrative two-group case where every nonelderly person is a worker, $N_W = N - N_E$. Start with

$$s_E = \frac{N_E}{N} \implies N_E = s_E N.$$

Therefore

$$N_W = N - s_E N = N(1 - s_E).$$

Substitute into $D = N_E / N_W$:

$$\begin{aligned} D &= \frac{s_E N}{N(1 - s_E)} \\ &= \frac{s_E}{1 - s_E}. \end{aligned}$$

Differentiate:

$$\boxed{\frac{dD}{ds_E} = \frac{1}{(1 - s_E)^2} > 0.}$$

This expression requires the simplifying identity $N_W = N - N_E$.

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Health-System Financial Sustainability

Let E_t denote health-care expenditure and R_t revenue assigned to health care during period t. A simple flow-budget sustainability condition is

$$R_t \geq E_t.$$

Define the financing gap:

$$\boxed{G_t = E_t - R_t.}$$

- $G_t > 0$: a financing deficit during this period.
- $G_t = 0$: current revenue exactly covers current expenditure.
- $G_t < 0$: a current surplus.

A single-year balance does not by itself establish long-run fiscal sustainability; demographic and medical spending trends also matter.

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Total Health Expenditure by Age Group

Let e_Y and e_E denote mean expenditure per person in the younger and elderly groups. Total expenditure is

$$E = e_Y N_Y + e_E N_E.$$

Since $s_E = N_E/N$, we have $N_E = s_E N$ and

$$N_Y = N - N_E = (1 - s_E)N.$$

Substitute:

$$\begin{aligned} E &= e_Y(1 - s_E)N + e_E s_E N \\ &= N\left[e_Y + (e_E - e_Y)s_E\right]. \end{aligned}$$

If $e_E > e_Y$ and all other factors are held fixed, composition-based aging raises E.

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Deriving the Expenditure Effect of Aging

Begin with

$$E(s_E) = N\left[e_Y + (e_E - e_Y)s_E\right].$$

For the partial derivative, hold total population and per-person spending in each age group fixed:

$$\begin{aligned}\frac{\partial E}{\partial s_E} &= N\left[\frac{\partial e_Y}{\partial s_E} + (e_E - e_Y)\frac{\partial s_E}{\partial s_E}\right] \\ &= N(e_E - e_Y),\end{aligned}$$

where e_Y and e_E are treated as constants in this partial derivative. Thus, if $e_E > e_Y$,

$$\boxed{\frac{\partial E}{\partial s_E} = N(e_E - e_Y) > 0.}$$

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Aging and Beveridge Financing

In a tax-funded system, denote health-care revenue by R.

$$G = E - R.$$

Substitute the age-composition expenditure function:

$$G = N\left[e_Y + (e_E - e_Y)s_E\right] - R.$$

Holding N, age-specific expenses, and assigned tax revenue R constant,

$$\begin{aligned}\frac{\partial G}{\partial s_E} &= \frac{\partial E}{\partial s_E} - \frac{\partial R}{\partial s_E} \\ &= N(e_E - e_Y) - 0.\end{aligned}$$

Consequently,

$$\boxed{e_E > e_Y \implies \frac{\partial G}{\partial s_E} > 0.}$$

Revenue or service reforms can alter this result in practice.

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Aging and Aggregate Health-Care Demand

Let h_Y and h_E be age-specific average health-care use, and suppose $h_E > h_Y$. Aggregate demand is

$$H_D = h_Y N_Y + h_E N_E.$$

Substitute $N_Y = (1 - s_E)N$ and $N_E = s_E N$:

$$\begin{aligned} H_D &= h_Y(1 - s_E)N + h_E s_E N \\ &= N\left[h_Y + (h_E - h_Y)s_E\right]. \end{aligned}$$

Holding other factors constant,

$$\boxed{\frac{\partial H_D}{\partial s_E} = N(h_E - h_Y) > 0.}$$

An older population uses more total care in this stylized scenario.

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Aging, Fixed Health Capacity, and Excess Demand

Suppose public health-care capacity is $\bar{H}$, unchanged as age composition shifts.

$$ED = H_D - \bar{H}.$$

Substituting the expression for age-specific demand:

$$ED = N\left[h_Y + (h_E - h_Y)s_E\right] - \bar{H}.$$

Differentiate while holding $\bar{H}$ fixed:

$$\begin{aligned}\frac{\partial ED}{\partial s_E} &= \frac{\partial H_D}{\partial s_E} - \frac{\partial \bar{H}}{\partial s_E} \\ &= N(h_E - h_Y).\end{aligned}$$

If $h_E > h_Y$,

$$\boxed{\frac{\partial ED}{\partial s_E} > 0.}$$

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Aging and the Waiting-Time Equilibrium

Let demand depend on waiting cost and population age composition:

$$H_D = H_D(C(w), s_E).$$

Assume

$$\frac{\partial H_D}{\partial C} < 0, \qquad \frac{\partial H_D}{\partial s_E} > 0, \qquad C'(w) > 0.$$

Given constant capacity $\bar{H}$, the equilibrium wait satisfies

$$\boxed{H_D(C(w^*), s_E) = \bar{H}.}$$

When aging raises demand at unchanged waiting time, the equilibrium adjusts through higher waiting cost if capacity and other demand determinants remain fixed.

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Step-by-Step: Differentiate the Waiting-Time Equation

Start with $H_D(C(w^*(s_E)), s_E) = \bar{H}$. Differentiate both sides with respect to s_E:

$$\frac{d}{ds_E} H_D(C(w^*(s_E)), s_E) = 0.$$

Apply the multivariable chain rule:

$$\frac{\partial H_D}{\partial C} \frac{dC(w^*)}{ds_E} + \frac{\partial H_D}{\partial s_E} = 0.$$

Apply the chain rule again:

$$\frac{dC(w^*)}{ds_E} = C'(w^*) \frac{dw^*}{ds_E}.$$

Substituting gives

$$\boxed{\frac{\partial H_D}{\partial C} C'(w^*) \frac{dw^*}{ds_E} + \frac{\partial H_D}{\partial s_E} = 0.}$$

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Step-by-Step: Sign of the Aging Effect on Waits

Starting with

$$H_{D,C}\,C'(w^*)\frac{dw^*}{ds_E} + H_{D,s_E} = 0,$$

move the demand-shift term to the other side:

$$H_{D,C}\,C'(w^*)\frac{dw^*}{ds_E} = -H_{D,s_E}.$$

Divide by $H_{D,C}\,C'(w^*) \neq 0$:

$$\boxed{\frac{dw^*}{ds_E} = -\frac{H_{D,s_E}}{H_{D,C}\,C'(w^*)} > 0.}$$

Why positive? $H_{D,s_E} > 0$ and $H_{D,C}\,C'(w^*) < 0$. The negative of a positive number divided by a negative number is positive.

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Beveridge Aging Mechanism: Summary

$$s_E \uparrow \implies \begin{cases} E \uparrow \implies G \uparrow & \text{if revenue is fixed,} \\ H_D \uparrow \implies ED \uparrow \implies w^* \uparrow & \text{if capacity is fixed.} \end{cases}$$

Interpretation

Aging can place pressure on tax financing and on scarce public treatment capacity. The magnitude depends on age-specific needs, taxes, capacity growth, prevention and technology.

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Bismarck Financing: Contributions and Benefits

Let N_W be the number of contributing workers, $\bar{y}$ the average contribution base per worker, and τ the health-insurance contribution rate. Revenue is

$$\boxed{R = \tau \bar{y} N_W.}$$

A balanced social insurance fund requires $R = E$, so

$$\tau \bar{y} N_W = E.$$

Divide both sides by $\bar{y} N_W > 0$:

$$\boxed{\tau^* = \frac{E}{\bar{y} N_W}.}$$

The required contribution rate grows with benefits expenditure and falls with the size of the contribution base.

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Deriving the Bismarck Contribution Rate Under Aging

Recall $E = e_Y N_Y + e_E N_E$. Insert this expression into the budget equation:

$$\tau \bar{y} N_W = e_Y N_Y + e_E N_E.$$

Dividing by $\bar{y} N_W$ yields

$$\boxed{\tau^* = \frac{e_Y N_Y + e_E N_E}{\bar{y} N_W}.}$$

Holding $N_W, \bar{y}, N_Y, e_Y, e_E$ fixed, a larger N_E raises the numerator:

$$\frac{\partial \tau^*}{\partial N_E} = \frac{e_E}{\bar{y} N_W} > 0.$$

If aging also shrinks the contributing worker population, the financing pressure can be stronger.

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Dependency Ratio and Contributions

For a simple illustration, suppose elderly medical expenditure dominates, so $E = e_E N_E$. The balanced-budget requirement is

$$\tau^* \bar{y} N_W = e_E N_E.$$

Rearrange:

$$\tau^* = \frac{e_E N_E}{\bar{y} N_W} = \frac{e_E}{\bar{y}} \frac{N_E}{N_W}.$$

Since $D = N_E / N_W$,

$$\boxed{\tau^* = \frac{e_E}{\bar{y}} D.}$$

Differentiate:

$$\boxed{\frac{\partial \tau^*}{\partial D} = \frac{e_E}{\bar{y}} > 0.}$$

This is an illustrative special case, not the general contribution formula.

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Bismarck Financing Gap Under Aging

With a fixed actual contribution rate τ, revenue is $R = \tau \bar{y} N_W$. The financing gap equals

$$G = E - R.$$

Substitute the age-composition formula:

$$G = N[e_Y + (e_E - e_Y)s_E] - \tau \bar{y} N_W.$$

If N, age-specific expenditure, τ, $\bar{y}$, and N_W are held fixed,

$$\boxed{\frac{\partial G}{\partial s_E} = N(e_E - e_Y) > 0.}$$

When N_W declines as s_E rises, revenue can fall as well, increasing the gap by an additional channel.

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Population Aging, Insurance, and Utilization

Let insured consumers pay the fraction α of the resource price p:

$$p_c = \alpha p, \qquad 0 \leq \alpha \leq 1.$$

Suppose aggregate health-care demand depends on the patient price and elderly share:

$$H_D = H_D(p_c, s_E).$$

Assume

$$\frac{\partial H_D}{\partial p_c} < 0, \qquad \frac{\partial H_D}{\partial s_E} > 0.$$

- Greater coverage reduces p_c and may raise demand for health care.
- Population aging shifts demand upward even when insurance terms remain unchanged.
- The two effects can reinforce pressures on the insurance fund.

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Bismarck Aging Mechanism: Summary

$$s_E \uparrow \implies E \uparrow, \quad H_D \uparrow .$$

If workers do not grow as fast as elderly beneficiaries, the dependency ratio may rise:

$$D = \frac{N_E}{N_W} \uparrow .$$

The budget balance contribution rate must then respond to changes in costs and contributors:

$$\tau^* = \frac{E}{\bar{y} N_W}.$$

Policy responses

Contribution-rate adjustment, broadening the contribution base, expenditure restraint, prevention, productivity improvements, and additional public financing.

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Health Policy: Chapter Synthesis

1. **All systems confront scarcity.** Public financing does not remove physical opportunity costs.
2. **Beveridge-type systems** often face difficult capacity and waiting-time choices.
3. **Bismarck-type systems** rely on risk pooling and must manage incentives and contribution adequacy.
4. **Insurance** protects against consumption risk but can generate selection and utilization effects.
5. **Population aging** may increase spending and challenge taxes, contributions and service capacity.

Next chapter

We now model *how infections emerge and move through the population*, creating demand for care that these health systems must finance and allocate.

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The SIR Model

Economic Epidemiology and Disease Control

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Why Economic Epidemiology Studies the SIR Model

An infectious disease can spread because one person's contact choices and infection status change other people's health risks.

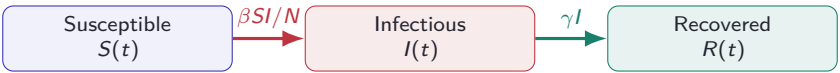
- The SIR model describes how an epidemic begins, rises, peaks and declines.
- It distinguishes people who are susceptible, currently infectious, and recovered.
- Health economists use it to study disease externalities, protection decisions, and surges in medical demand.

Central question

How can infectious cases rise rapidly at first and later decline even when there is no permanent lockdown or universally effective cure?

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The Susceptible–Infectious–Recovered Flow



Assume a closed population with no births, deaths or reinfections, and lasting immunity after recovery.

$$\boxed{N = S(t) + I(t) + R(t).}$$

People change groups through infection and recovery; the total population remains fixed.

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Understanding the Three Population Compartments

Susceptible $S(t)$: People who can acquire the infection. Without vaccination, this stock falls when new infections occur.

Infectious $I(t)$: People who currently transmit the disease. This stock rises with new infections and falls with recoveries.

Recovered $R(t)$: People who have left the infectious compartment and, in the basic model, cannot immediately become susceptible again.

$$\begin{aligned} \text{Change in } S &= -\text{New infections}, \\ \text{Change in } I &= \text{New infections} - \text{Recoveries}, \\ \text{Change in } R &= \text{Recoveries}. \end{aligned}$$

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Deriving New Infections from Contacts

Let c be the number of potentially infectious contacts per susceptible person per unit time, and let p be the infection probability given a contact with an infectious person. Under random mixing:

1. The fraction of contacts with infectious persons is I/N.
2. The expected number of infections per susceptible person is $cp(I/N)$.
3. Multiplying by the S susceptible people yields total new infections.

$$\text{New infections per unit time} = S\, c\, p \frac{I}{N}.$$

Define $\beta = cp$. Therefore,

$$\boxed{\text{New infections} = \beta \frac{SI}{N}.}$$

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Deriving the Recovery Flow

Let $\gamma > 0$ be the per-person recovery (or removal) hazard per unit time.

- During a short interval Δt, each infectious person recovers with probability approximately $\gamma \Delta t$.
- For I infectious persons, the expected number recovering in that interval is approximately $\gamma I \Delta t$.
- Divide by Δt: recoveries per unit time are γI.

$$\boxed{\text{Recoveries per unit time} = \gamma I.}$$

For a constant recovery hazard, expected duration of infectiousness is $1/\gamma$. For example, $\gamma = 0.1$ per day implies an average duration of ten days.

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Derivation: Three SIR Differential Equations

For a short time interval Δt,

$$S(t+\Delta t) - S(t) \approx -\beta \frac{SI}{N} \Delta t,$$
$$I(t+\Delta t) - I(t) \approx \left(\beta \frac{SI}{N} - \gamma I\right) \Delta t,$$
$$R(t+\Delta t) - R(t) \approx \gamma I \Delta t.$$

Divide each equation by Δt and let $\Delta t \to 0$:

$$\boxed{\frac{dS}{dt} = -\beta \frac{SI}{N}, \quad \frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I, \quad \frac{dR}{dt} = \gamma I.}$$

These equations describe continuous-time changes in the three stocks.

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Verifying Population Conservation

The population identity is $N = S + I + R$. Differentiate with respect to time:

$$\frac{dN}{dt} = \frac{dS}{dt} + \frac{dI}{dt} + \frac{dR}{dt}.$$

Substitute the SIR differential equations:

$$\begin{aligned}\frac{dN}{dt} &= -\beta\frac{SI}{N} + \left(\beta\frac{SI}{N} - \gamma I\right) + \gamma I \\ &= -\beta\frac{SI}{N} + \beta\frac{SI}{N} - \gamma I + \gamma I \\ &= 0.\end{aligned}$$

Thus, the model changes the population's composition without changing its total size.

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Graphical Evolution of an Epidemic

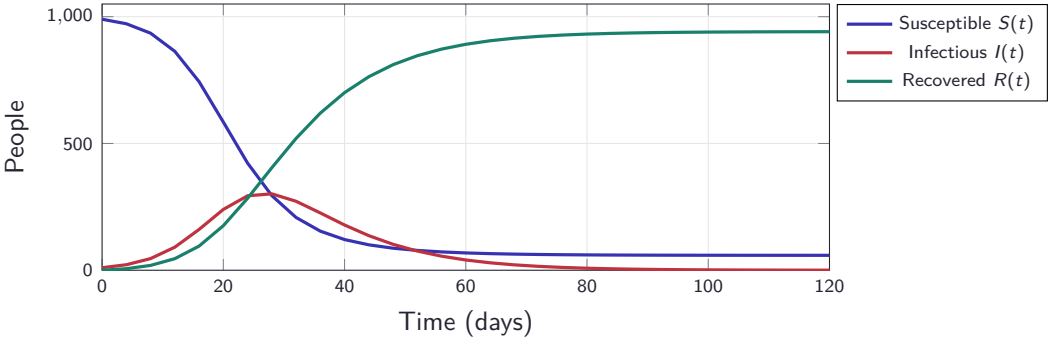


Illustration: $N = 1000$, $S(0) = 990$, $I(0) = 10$, $R(0) = 0$, $\beta = 0.30$, $\gamma = 0.10$ per day.

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Interpreting the SIR Curves

1. **Susceptible:** declines if $S > 0$, $I > 0$ and $\beta > 0$ because $S' = -\beta SI/N < 0$.
2. **Infectious:** initially rises when new infections exceed recoveries; later falls when recoveries dominate.
3. **Recovered:** increases monotonically because $R' = \gamma I \geq 0$.

$$\underbrace{\frac{dI}{dt}}_{\text{Net change in infectious}} = \underbrace{\beta \frac{SI}{N}}_{\text{New infections}} - \underbrace{\gamma I}_{\text{Recoveries}}.$$

Key idea

The epidemic peaks when two *flows* become equal, not when transmission stops.

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Basic Reproduction Number $\mathcal{R}_0$

Suppose nearly everybody is initially susceptible, so $S \approx N$. Each infectious person generates infection at approximate rate β and remains infectious for an average $1/\gamma$ time units. Thus, the basic reproduction number is

$$\boxed{\mathcal{R}_0 = \beta\left(\frac{1}{\gamma}\right) = \frac{\beta}{\gamma}.}$$

- $\mathcal{R}_0 > 1$: an introduced infection can initially grow.
- $\mathcal{R}_0 = 1$: initial transmission is at the threshold.
- $\mathcal{R}_0 < 1$: a small introduction tends to decline.

For $\beta = .30$ and $\gamma = .10$, $\mathcal{R}_0 = 3$.

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Effective Reproduction Number $\mathcal{R}_t$

At time t, only fraction $S(t)/N$ remains susceptible.

$$\boxed{\mathcal{R}_t = \frac{\beta}{\gamma}\frac{S(t)}{N} = \mathcal{R}_0\frac{S(t)}{N}.}$$

Rearrange the infectious differential equation:

$$\begin{aligned}\frac{dI}{dt} &= \beta\frac{SI}{N} - \gamma I \\ &= \gamma I\left(\frac{\beta S}{\gamma N} - 1\right) \\ &= \gamma I(\mathcal{R}_t - 1).\end{aligned}$$

For $I(t) > 0$, the sign of $I'(t)$ is determined by $\mathcal{R}_t - 1$.

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Deriving the Epidemic Peak Condition

At an interior peak of the infectious stock $I(t)$, its derivative is zero:

$$\frac{dI}{dt} = 0.$$

Substitute the infectious equation:

$$\beta \frac{SI}{N} - \gamma I = 0.$$

For $I > 0$, divide by I:

$$\frac{\beta S}{N} - \gamma = 0.$$

Rearrange and substitute $\mathcal{R}_0 = \beta/\gamma$:

$$\boxed{\frac{S_{\text{peak}}}{N} = \frac{\gamma}{\beta} = \frac{1}{\mathcal{R}_0}.}$$

The infectious stock begins to decline after susceptible people become too scarce for each infection to replace itself.

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Herd-Immunity Threshold in the Simple Model

At the point where $\mathcal{R}_t = 1$, the susceptible share is

$$\frac{S}{N} = \frac{1}{\mathcal{R}_0}.$$

Therefore the complementary immune/removed fraction under the model's assumptions is

$$\boxed{H_c = 1 - \frac{1}{\mathcal{R}_0}.}$$

For $\mathcal{R}_0 = 3$:

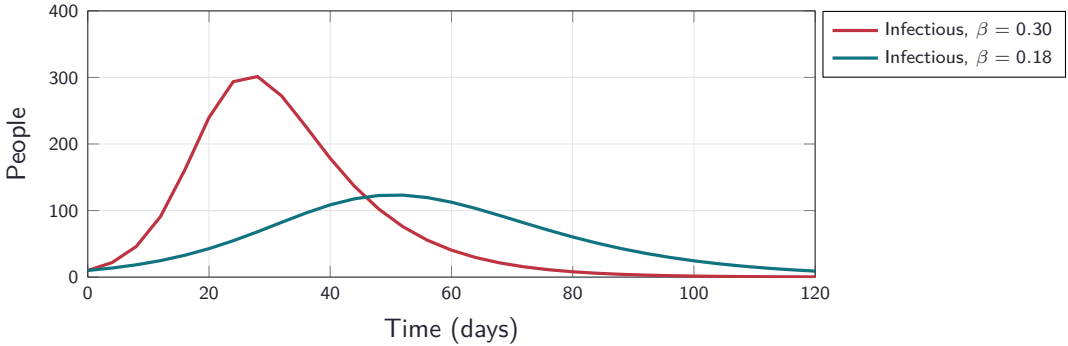
$$H_c = 1 - \frac{1}{3} = \frac{2}{3} \approx 66.7\%.$$

Limits of the interpretation

This is a homogeneous-mixing threshold, not an automatic vaccine target. Reaching the threshold starts the decline; it does not immediately end all new infections (epidemic overshoot is possible).

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Disease Control: Flattening the Infectious Curve



Same initial populations; $\gamma = 0.10$ per day. Lower transmission reduces and delays the peak in this example.

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A Fully Worked SIR Numerical Example

Assume $N = 1000$, $S = 990$, $I = 10$, $R = 0$, $c = 5$ contacts per day, $p = .06$ per infectious contact and $\gamma = .10$ per day.

$$\beta = cp = 5(0.06) = 0.30,$$

$$\text{New infections/day} = .30\left(\frac{990 \times 10}{1000}\right) = 2.97,$$

$$\text{Recoveries/day} = .10(10) = 1.00,$$

$$\frac{dI}{dt} = 2.97 - 1.00 = 1.97 \text{ people/day}.$$

Also, $S' = -2.97$, $R' = 1.00$, $\mathcal{R}_0 = 3$, and $\mathcal{R}_t = 3(990/1000) = 2.97$ initially.

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Why Behavior Changes Transmission

Transmission depends on the frequency and risk of social interaction.

$$\beta = c\,p.$$

- Fewer risky contacts reduce c.
- Protective measures can lower transmission probability p per contact.
- Either mechanism reduces $\beta SI/N$ for the same population stocks.

Preventive behavior can impose costs: lost income, reduced mobility, foregone social interaction, or government spending.

Economic problem

Prevention requires comparing expected infection losses with the opportunity cost of protection and reductions in contact.

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Infectious-Disease Externalities

An individual's protective action often benefits people beyond the decision-maker.

- **Private benefit:** a lower personal probability of infection.
- **External benefit:** fewer transmissions to other people.
- **Social benefit:** private plus external risk reduction.

If prevention a yields private marginal benefit $MPB(a)$ and external marginal benefit $MEB(a)$, then

$$MSB(a) = MPB(a) + MEB(a).$$

A social optimum equates $MSB(a) = MC(a)$, while private choice often satisfies only $MPB(a) = MC(a)$.

Policy implication

Information, vaccination support, or targeted public-health measures may improve welfare when transmission externalities are important.

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Infectious Disease and Hospital Capacity

Suppose κ is a simplified ratio of required hospital places to infectious people, and K is medical capacity.

$$B(t) = \kappa I(t).$$

Capacity is exceeded if

$$\boxed{\kappa I(t) > K.}$$

Holding K and κ constant, an intervention that lowers the peak of $I(t)$ can also reduce the peak of required hospital places.

Interpretation

This proportional relationship is a teaching approximation, not a clinical forecast: true beds required depend on severity, admission delays, length of stay, age, and treatment effectiveness.

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SIR and Health-Policy Pressures

	Beveridge-type pressures	**Bismarck-type pressures**
Sharp epidemic wave	Hospitals and budgets become congested; waiting times may increase	Medical claims, insurer payments and risk-pool expenses may rise
Disease control	Lower peak demand helps scarce service capacity	Lower expected claims may relieve fund spending
Policy instruments	Capacity investment, priority, prevention	Prevention, insurance funding, provider payment

$$\text{Disease dynamics} \Longrightarrow \text{Health-care demand} \Longrightarrow \text{Economic pressure.}$$

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Limits and Lessons of the SIR Model

The basic SIR model assumes random mixing, constant parameters, immediate infectiousness, no demographic turnover, no reinfections and lasting immunity.

- Real transmission differs by age, contacts, workplace, behavior and immunity.
- Other models add exposure periods (SEIR), vaccination, waning immunity, and time-varying contact rates.
- Estimates of parameters and hospital burdens require reliable real-world evidence.

What we have learned

The epidemic's turning point is explained by the balance between infection and recovery flows; preventive choices can change this balance and thereby medical demand.

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Pharmaceutical Innovation

Drug Development, Patents, Regulation, and Technology

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Pharmaceutical Innovation and Medical Technology

Several questions guiding this chapter

How do medicines move from a scientific idea to patients? Why are drug development and innovation costly, risky, and regulated? Why can medical technology improve health while raising spending?

1. The life cycle of a drug and uncertainty in development.
2. Patents, market exclusivity, and incentives for innovation.
3. Induced innovation and demand-driven pharmaceutical research.
4. Regulation of pharmaceutical products and markets.
5. Technology, health-care prices, and medical expenditures.

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Why Is the Pharmaceutical Industry Economically Different?

A medicine cannot usually be judged before it has been thoroughly studied. This creates a distinctive combination of high initial expenditure, uncertain success, and patient safety concerns.

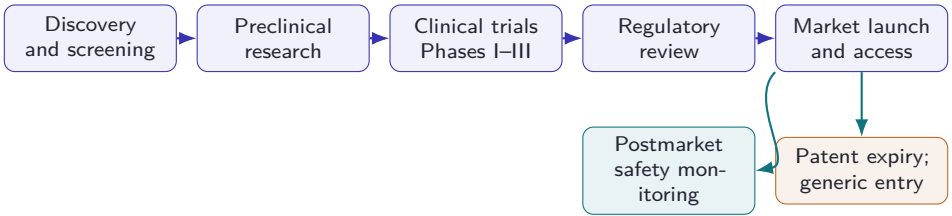
- **Large fixed costs:** discovery, experiments, trials, data analysis, and approval.
- **Uncertain benefits:** a promising molecule may be ineffective or unsafe.
- **Low or variable marginal costs:** producing an additional pill may cost much less.
- **Public interest:** drug quality and safety are not fully observable to patients.

Economic tension

Society wants both incentives to discover new medicines and affordable access to medicines after they exist.

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The Life Cycle of a Drug: The Big Picture



- The stages are a *sequence of tests*: failure can occur before any sale is made.
- Drug research continues after approval through safety monitoring and sometimes additional studies.
- Commercial experience later changes when competing generic or biosimilar products become available.

Source for regulatory stages: U.S. FDA, *The Drug Development Process*.

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Stage 1: Discovery and Early Research

The initial objective is to identify a biological target and a candidate substance that might prevent or treat disease.

- Researchers investigate the mechanism of disease and search for possible therapeutic compounds.
- Laboratory screening identifies candidates with potentially useful biological effects.
- Promising compounds are optimized for potency, stability, and tolerability.

Economic interpretation

Money and scientific effort are invested **before there is evidence that a marketable drug exists**. Research produces information and possibilities, rather than immediate revenue.

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Stage 2: Preclinical Research

Before routine testing in people, developers assess a candidate through laboratory investigations and, when required and appropriate, animal studies.

- Study how the drug acts and is processed in the body.
- Investigate possible toxicity and adverse effects.
- Decide whether the potential benefit justifies human testing.

Why does this stage matter? Early safety screening can prevent exposing trial volunteers to avoidable risks, even though preclinical results do not guarantee human safety or effectiveness.

Decision under uncertainty

Proceed to clinical development only when the expected benefits of learning more justify the additional expected costs and risks.

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Stage 3: Clinical Trials and Their Purposes

Stage	Primary question	Economic meaning
Phase I	How safe and tolerable is the drug, and how does the body handle it?	First major clinical uncertainty is reduced.
Phase II	Does it show promising activity or benefit in the target patients?	Evidence helps determine whether to undertake larger trials.
Phase III	Does it provide sufficient evidence of effectiveness and safety in larger groups?	Often a particularly expensive investment before filing for approval.

Trials are not identical for every product; designs and regulatory pathways can vary. In some circumstances, approval may occur without the conventional sequence.

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Stage 4: Approval, Marketing, and Continued Monitoring

- A regulator evaluates evidence on benefits, risks, manufacturing quality, and proposed labeling.
- In the United States, the FDA makes a marketing-approval decision for drugs within its jurisdiction.
- Approval does **not** prove that a drug is completely risk-free or inexpensive.
- After launch, surveillance and studies may detect rare or long-term problems that earlier trials could not detect.

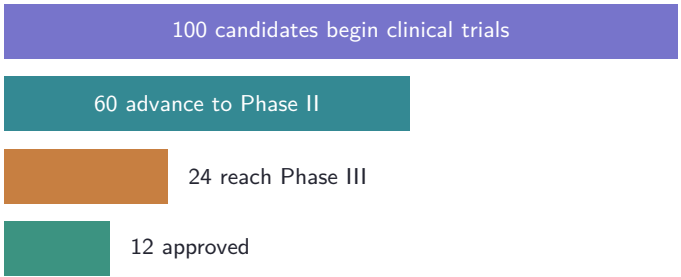
Three distinct economic questions

Regulatory approval: Is the benefit–risk evidence sufficient?
Pricing and coverage: What will buyers pay, and who bears the cost?
Cost effectiveness: Are the extra health gains worth the extra resources?

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Why Do So Many Drug Candidates Disappear?

Illustrative attrition funnel: The number of candidates typically becomes smaller as development advances. The counts below are an illustrative teaching example, not a universal success rate.



Implication: Developing several medicines that fail may be part of the cost of obtaining one successful medicine. Actual rates differ by cohort, disease area, and period.

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The Sources of Uncertainty in Drug Development

Drug development has **several separate uncertainties**, not a single known outcome.

1. **Scientific uncertainty:** will the compound affect the intended biological target?
2. **Clinical uncertainty:** will patients benefit enough, and will side effects be tolerable?
3. **Regulatory uncertainty:** will the evidence support approval for a proposed indication?
4. **Commercial uncertainty:** will doctors prescribe it, payers cover it, and patients use it?
5. **Competitive uncertainty:** will a rival product arrive first or perform better?

Research and trials can reduce some uncertainties but cannot remove all future risks.

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The Economic Costs of Drug Development

An accurate cost concept should distinguish several categories:

- **Direct spending:** laboratory work, staff, trials, monitoring, manufacturing development, and applications.
- **Failed-project spending:** money invested in candidates that never reach the market.
- **Cost of capital:** research funds are tied up for years before any revenue may arrive.
- **Postapproval spending:** safety monitoring, additional studies, and production-scale improvements.

Important qualification

There is no single development cost that describes every drug. Reported estimates depend on research methods, which costs are included, and the drugs in the sample.

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A Portfolio Example: Why Failures Raise Cost

Suppose a hypothetical research program starts with 100 clinical candidates. Costs for each candidate *entering* each phase are shown below.

Phase	Number entering	Cost each (million $)	Total (million $)
I	100	10	1,000
II	60	30	1,800
III	24	100	2,400
Total			**5,200**

If 12 products are approved, then the average clinical-phase expenditure across the portfolio is

$$\boxed{\frac{5{,}200}{12} = \$433.33 \text{ million per approved drug.}}$$

This is only an illustrative calculation; it excludes discovery, preclinical costs, capital costs, and postapproval spending.

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Mathematics of Success Probability

Let q_1, q_2, and q_3 be the *conditional* probabilities of passing successive stages, including the final decision within the third-stage outcome.

$$\Pr(\text{success}) = \Pr(A_1)\Pr(A_2 \mid A_1)\Pr(A_3 \mid A_1, A_2).$$

Writing these conditional probabilities as q_1, q_2, q_3 gives

$$\boxed{p = q_1 q_2 q_3}.$$

For an illustrative set of probabilities,

$$p = (0.60)(0.40)(0.50) = 0.12.$$

Thus, starting with 100 candidates, the expected number eventually approved is

$$\boxed{100 \times 0.12 = 12}.$$

These probability values are hypothetical and should not be interpreted as guaranteed success rates.

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Expected R&D Cost Per Successful Drug

Suppose the three clinical stages cost c_1, c_2, c_3 for candidates entering them, and stage progression is determined by conditional probabilities q_1, q_2, q_3.

$$\mathbb{E}[C \text{ per starting candidate}] = c_1 + q_1 c_2 + q_1 q_2 c_3,$$
$$p = q_1 q_2 q_3.$$

Under a large, homogeneous portfolio and these assumptions,

$$\boxed{\text{Expected portfolio spending per success} = \frac{c_1 + q_1 c_2 + q_1 q_2 c_3}{q_1 q_2 q_3}.}$$

Interpretation: When success probability falls, the cost allocated to each eventual successful drug generally rises, holding stage costs and the relevant progression pattern fixed.

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Why the Time Value of Money Matters

A dollar spent today could have earned a return if invested elsewhere. Drug revenues may arrive only after years of research, trials, and approval.

$$PV(\text{future revenue}) = \frac{F}{(1+r)^T}$$

Here F is a future revenue receipt, r is the annual discount rate, and T is the number of years until receipt. **Illustration:** For $F = \$100$ million received in ten years at $r = 0.05$,

$$PV = \frac{100}{1.05^{10}} \approx \$61.4 \text{ million}.$$

Meaning

A project that generates substantial revenue far in the future can still be unattractive today if success is uncertain and upfront investment is very high.

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What Is a Pharmaceutical Patent?

A patent is a legal right that can prevent others from commercially practicing the claimed invention without authorization during its enforceable term.

- It can protect a drug molecule, manufacturing method, formulation, or qualifying use, depending on the claims.
- It can help an innovator capture some of the returns from costly research.
- It also temporarily constrains certain forms of competition, affecting prices and access.

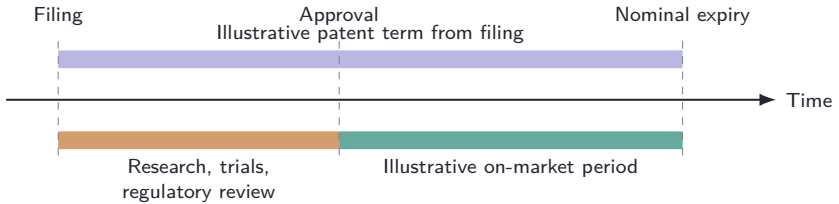
Patent versus FDA exclusivity

A **patent** is granted by the patent office. **Regulatory exclusivity** is a separate legal restriction on certain competing approvals. Their periods can overlap, differ, or be absent.

U.S. reference: FDA, *Frequently Asked Questions on Patents and Exclusivity*.

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A Patent Timeline: Filing Is Not Market Entry



In the United States, the ordinary patent term is generally **20 years from the relevant filing date**, subject to legal qualifications. A drug may spend many of those years in development, rather than on the market.

Important

Actual effective protection depends on patent scope, filing dates, patent-term adjustments or extensions, litigation, and separate regulatory exclusivities.

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Why Patents Can Encourage Pharmaceutical Innovation

Consider a firm deciding whether to pay a fixed R&D cost F to develop a new medicine.

- Without effective protection, competitors may imitate an invention and reduce the original firm's expected profits.
- With temporary protection, an innovator may be able to recover more of its initial investment.
- Protection can therefore increase the expected return to R&D and make more projects financially attractive.

$$\boxed{\text{Invest if } \mathbb{E}[PV(\text{future profit})] \geq F.}$$

Trade-off: Stronger expected returns may promote future discoveries, but higher present prices can restrict access to existing treatments.

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Patent Protection and Monopoly Pricing: A Simple Model

Suppose inverse market demand for a protected medicine is $P(Q) = a - bQ$, with $a > c > 0$, $b > 0$, and constant marginal production cost c.

$$TR(Q) = P(Q)Q = aQ - bQ^2, \qquad MR(Q) = a - 2bQ.$$

A single profit-maximizing seller sets $MR = MC = c$:

$$a - 2bQ_M = c \quad \Rightarrow \quad \boxed{Q_M = \frac{a-c}{2b}}.$$

Substitute into demand:

$$\boxed{P_M = \frac{a+c}{2}}.$$

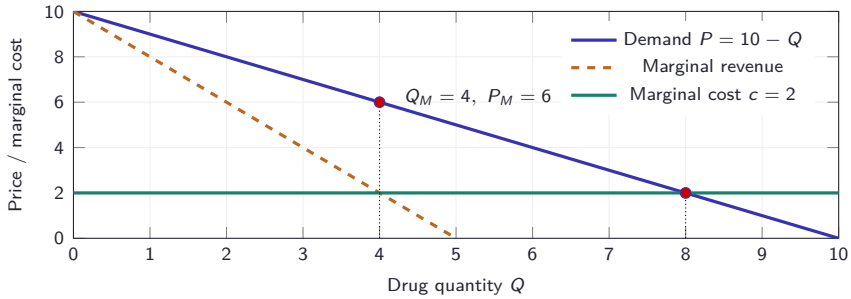
Under the simple competitive benchmark $P_C = c$,

$$\boxed{Q_C = \frac{a-c}{b} > Q_M}.$$

Interpretation: Market power raises price and lowers quantity relative to this benchmark. This abstracts from bargaining, regulation, and insurance.

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The Patent Trade-off in a Demand Diagram



The monopolist restricts output relative to the price-equals-marginal-cost benchmark. The loss in gains from trade is **static deadweight loss**; the possible reward for innovation is a **dynamic benefit**.

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What Happens When Generic Competition Arrives?

Generic entry, when legally permitted and successfully approved, can erode a brand producer's market power.

- Multiple suppliers may offer equivalent small-molecule medicines.
- Competition and lower upfront development costs can reduce prices.
- Buyers, insurers, and public programs may save money or treat more patients.

Economic interpretation

Patent and exclusivity rules affect the **timing of competition**. Once competition becomes effective, price can move closer to production and distribution costs, though actual markets are more complicated than perfect competition.

Important distinction: Biologic medicines typically face *biosimilar*, rather than conventional generic, competition.

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Induced Innovation: Why Demand Shapes Discovery

Induced innovation means the direction and intensity of technical progress respond to economic incentives.

- A larger patient population may promise greater sales of an effective treatment.
- Greater insurance coverage, willingness to pay, or public procurement can enlarge expected demand.
- Firms may redirect research toward conditions offering higher expected returns.

$$\boxed{\text{Larger potential market} \longrightarrow \text{Higher expected return} \longrightarrow \text{More R\&D incentives}}$$

This is a tendency, not a guarantee: scientific feasibility, research costs, policies, and competition also matter.

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A Simple Mathematical Model of Induced Innovation

Suppose developing a medicine costs $F > 0$. Let M be the number of potential patients, $m > 0$ the expected contribution margin per patient, and $p \in (0, 1]$ the probability of commercial success.

$$\boxed{\mathbb{E}[\Pi] = pMm - F}.$$

Differentiate with respect to market size:

$$\boxed{\frac{\partial \mathbb{E}[\Pi]}{\partial M} = pm > 0}.$$

A project has nonnegative expected net return if

$$pMm \geq F \quad \Longleftrightarrow \quad \boxed{M \geq M^* = \frac{F}{pm}}.$$

Meaning: A larger market can induce a project that would have been unprofitable at smaller scale. This is a stylized model, not a prediction for an individual drug.

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Market Size and the Incentive to Innovate: Graph

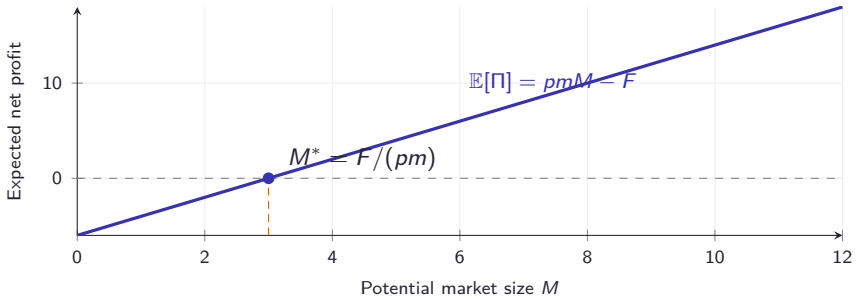


Illustration: Here $pm = 2$ and $F = 6$, so the investment breaks even at $M^* = 3$ in arbitrary units. Above that market size, expected net return becomes positive.

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Induced Innovation and Changes in Disease Burden

Imagine a disease that becomes more common because the population is aging.

1. More patients may need treatment for that disease.
2. The potential market for relevant medicines becomes larger.
3. Expected revenue may increase, holding prices, costs, and success probabilities fixed.
4. More firms may invest in therapies, diagnostic methods, or delivery systems.

Important distinction

The disease with the *greatest social health burden* need not offer the *highest private profit*. Low-income populations and rare conditions can therefore be underserved without additional policies.

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Private and Social Incentives for Medical Research

A firm weighs private revenues against private R&D expenditures. Society also values improvements in survival, pain, productivity, and spillover knowledge.

$$\boxed{\text{Private return} \neq \text{social return in general.}}$$

- **Positive spillovers:** scientific knowledge may help future researchers.
- **Unmet need:** social benefits may be large even when patient purchasing power is small.
- **Access concerns:** private pricing can exclude some patients whose treatment benefit exceeds marginal cost.

Public research grants, prizes, targeted incentives, and procurement can change which medicines are pursued.

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Why Regulate the Pharmaceutical Industry?

Medicines create serious information problems: patients and often prescribers cannot independently verify quality, efficacy, or long-term safety.

- **Safety:** detect and limit unacceptable risks.
- **Effectiveness:** require adequate evidence of clinical benefit.
- **Quality:** enforce consistent manufacturing and labeling standards.
- **Information:** regulate promotional and safety claims to reduce misleading messages.
- **Competition and access:** other government policies can address patents, purchasing, reimbursement, and prices.

Note: The U.S. FDA's product-approval responsibility is distinct from most drug-price and insurance-coverage decisions.

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Regulation as a Balance of Benefits and Costs

An effective regulatory system protects patients while avoiding needless barriers to valuable treatment.

Potential gains from oversight	Potential costs of oversight
Earlier detection of harmful side effects	More trials and documentation expense
Better evidence of medical benefit	Longer time before some treatments become available
Manufacturing quality and reliable information	Higher fixed compliance costs; entry barriers
Public trust and more informed decisions	Regulatory delay can carry health opportunity costs

Policy goal: obtain credible clinical evidence and acceptable quality at an appropriate social cost; neither zero regulation nor maximum delay is automatically optimal.

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Regulation Continues After Approval

Clinical trials cannot identify every rare side effect or long-term interaction, especially for groups that were less represented in the trials.

- Safety reports and observational evidence can reveal new risks.
- Regulators may update warnings, restrict indications, require new studies, or in some cases withdraw approval.
- Developers may study additional indications, formulations, and patient groups.

Health-economics lesson

Approval is a **decision based on available evidence**, not the end of learning. Better postmarket information may raise or lower a drug's assessed clinical and economic value.

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Medical Technology: What Does the Term Include?

Medical technology is wider than pharmaceuticals. It includes

- drugs and vaccines;
- imaging, diagnostic tests, and monitoring devices;
- surgical tools, prosthetics, and therapeutic equipment;
- digital health systems and new methods of delivering care.

Technology can change **which conditions are treatable**, **how much health gain is achievable**, **how many patients use care**, and **how much each episode costs**.

Central question

When care becomes technologically better, does the *price per service*, the *cost per health gain*, or *total national spending* rise or fall?

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Technology and the Price of Health Care

Three different price concepts must not be confused.

1. **Price per item or service:** payment for one drug course or procedure.
2. **Cost per successfully treated patient:** includes effectiveness and any avoided follow-up treatment.
3. **Cost per unit of health improvement:** accounts for differences in health gains and quality.

A new treatment can have a **higher sticker price** yet achieve greater health improvements or reduce other medical costs.

Interpretation

Technology does not have a universally positive or negative effect on quality-adjusted treatment prices.

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An Example of Quality-Adjusted Treatment Cost

Imagine two treatments with different costs and health benefits. These are artificial numbers for explanation.

Treatment	Resource cost ($)	Health gain (QALY)
Older treatment	100	0.5
Newer treatment	150	1.5

$$\text{Cost per QALY: old} = \frac{100}{0.5} = \$200, \qquad \text{new} = \frac{150}{1.5} = \$100.$$

Despite its higher total price, the newer treatment costs *less per unit of health gain* in this simplified comparison. **Caution:** Incremental cost effectiveness is the preferred measure when treatments are alternatives (discussed later).

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Technology Can Save Costs or Expand Treatment

Cost-saving innovations

- A cheaper diagnostic test with equal accuracy.
- A medicine that prevents expensive complications.
- A less resource-intensive procedure with similar outcomes.

Treatment-expanding innovations

- A therapy for a previously untreatable disease.
- More repeated or intensive monitoring.
- A safer procedure used by many additional patients.

Conclusion: A technology can be clinically valuable while increasing total medical expenditure because it expands the number or intensity of treatments.

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Understanding Medical Expenditure: Price Times Quantity

For a single well-defined service,

$$\boxed{E = P \times Q},$$

where E is spending, P is expenditure per unit of service, and Q is the number of services. If there are multiple technologies or treatments, total spending is

$$\boxed{E = \sum_{j=1}^{J} P_j Q_j}.$$

Changes in technology can therefore affect spending through **prices**, **treatment frequency**, **intensity**, and the **mix of services**.

Economic interpretation

Total expenditure can rise even if the cost of a particular service falls, provided sufficiently many additional services are supplied or demanded.

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Deriving the Growth of Medical Expenditure

For one service, start with

$$E = PQ.$$

Differentiate with respect to time using the product rule:

$$\frac{dE}{dt} = Q\frac{dP}{dt} + P\frac{dQ}{dt}.$$

Divide both sides by $E = PQ$:

$$\frac{1}{E}\frac{dE}{dt} = \frac{1}{P}\frac{dP}{dt} + \frac{1}{Q}\frac{dQ}{dt}.$$

Thus the **instantaneous percentage growth rate** is

$$\boxed{\frac{d\ln E}{dt} = \frac{d\ln P}{dt} + \frac{d\ln Q}{dt}.}$$

This exact decomposition holds for a fixed single service; many-service analysis must also consider changing treatment mix.

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A Numerical Example: Why Spending Can Rise

	Before innovation	After innovation
Price per treatment ($)	100	80
Number treated	1,000	1,500
Total spending ($)	100,000	120,000

$$E_0 = 100 \times 1{,}000 = 100{,}000, \qquad E_1 = 80 \times 1{,}500 = 120{,}000.$$

Price fell by 20%, but quantity rose by 50%, so total spending increased by 20%.

Central lesson

A lower price per service does not guarantee lower total spending. Expansion in utilization can outweigh a reduction in unit cost.

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Graph: Lower Unit Price, Higher Total Expenditure

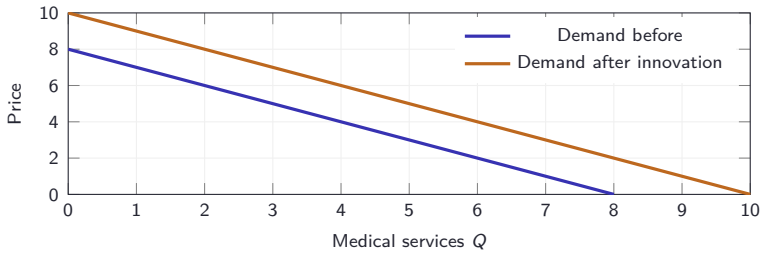


The bars show total expenditure from the preceding example, *not* a measured historical trend. More people receiving the less expensive treatment raises the total health-care budget.

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How Technology Can Shift Health-Care Demand

When a new procedure is safer, less painful, or effective for more conditions, more patients may want it at any given price.



Meaning: Technology may stimulate demand as well as change supply. If insurance covers the service, its out-of-pocket price may further encourage use.

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Why Insurance Matters for Technology Adoption

Insurance can help people access costly medical innovations by spreading expenditure across many contributors.

- Coverage lowers the patient's immediate out-of-pocket price and can increase use.
- Providers may adopt new equipment or procedures when reimbursement is adequate.
- Patients and physicians may not face the full social resource cost at the point of treatment.

$$\boxed{P_{\text{patient}} = \alpha P, \qquad 0 \leq \alpha \leq 1}.$$

A smaller coinsurance fraction α can increase demand if demand decreases with the price patients pay. **Qualification:** This does not imply that every additional use is wasteful; some previously untreated patients may gain valuable care.

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Measuring Value: Incremental Cost-Effectiveness

When a new therapy replaces an old one, evaluate the *change* in resource cost against the *change* in health benefit.

$$\boxed{ICER = \frac{C_{\text{new}} - C_{\text{old}}}{H_{\text{new}} - H_{\text{old}}}}, \quad H_{\text{new}} > H_{\text{old}}.$$

Using the earlier example,

$$\Delta C = 150 - 100 = 50, \qquad \Delta H = 1.5 - 0.5 = 1.0 \text{ QALY}.$$

Hence,

$$\boxed{ICER = \$50 \text{ per additional QALY}.}$$

Interpretation

The ICER indicates the additional spending needed for one additional health unit. Whether that is worthwhile depends on the opportunity cost of health-care resources and the decision context.

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Cost-Effective Does Not Necessarily Mean Affordable

Even a medicine that generates large health gains for its price may have a substantial **budget impact** if millions of people become eligible.

$$\boxed{\Delta B \approx N_{\text{eligible}} \times u \times (C_{\text{new}} - C_{\text{old}})},$$

where u is the treated share, holding other costs and offsets fixed. **Illustration:** If one million patients are eligible, half receive treatment, and extra net cost is 200 per patient,

$$\Delta B \approx 1{,}000{,}000 \times 0.5 \times 200 = \$100 \text{ million}.$$

Difference

Cost effectiveness asks whether health benefits justify additional resources; **budget impact** asks how much funding is required in total.

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Technology and the Rise in Medical Expenditures

Several mechanisms connect technical progress to aggregate medical spending:

1. **New treatment possibilities:** patients with previously untreatable conditions receive care.
2. **Greater treatment intensity:** patients receive additional procedures, tests, or drug combinations.
3. **Higher or lower unit costs:** a new service may be costly, while automation may make another cheaper.
4. **Population effects:** more people may survive long enough to need further care.
5. **Insurance and payment rules:** reimbursement can affect diffusion and utilization.

The net effect on expenditure is empirical and depends on the technologies and policy environment.

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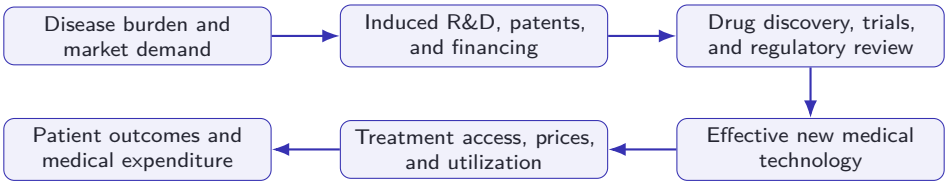
Policy Responses: Encouraging Value and Access

- **Public support for research:** fund socially valuable knowledge and neglected conditions.
- **Patents with time limits:** reward inventions while permitting later competition.
- **Rigorous approval and monitoring:** improve evidence on benefit, risk, and quality.
- **Generic and biosimilar competition:** expand affordability where safe and legally permitted.
- **Health technology assessment:** compare incremental benefits, costs, and opportunity costs.
- **Purchasing and reimbursement design:** reward clinical value and reduce low-value use.

No one policy removes the fundamental tension among innovation, access, safety, and expenditure.

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Synthesis: Connecting the Seven Topics



Central conclusion

Scientific innovation can increase survival and quality of life, but its cost and diffusion create difficult choices about incentives, equitable access, patient safety, and the sustainable financing of health care.

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Chapter Review Questions

1. Explain why the life cycle of a drug contains repeated decisions to continue or abandon development.
2. Show how uncertainty and unsuccessful candidates raise average research cost per approved drug.
3. Explain the difference between a patent, regulatory exclusivity, and generic entry.
4. Derive the privately profitable market-size threshold for a pharmaceutical research project.
5. Why can a medically beneficial technology have a higher price but a lower cost per health gain?
6. Starting with $E = PQ$, explain how spending can rise even while P falls.
7. Distinguish regulatory approval, cost effectiveness, and budget affordability.

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Notes: Institutional details use U.S. examples; patent terms, approval routes, and reimbursement policy vary by country. All classroom numerical models and diagrams are illustrative.

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Integrated Course Review

1. **Health Policy:** financing and allocation mechanisms determine how care is paid for and distributed under scarcity.
2. **SIR:** disease dynamics create transmission externalities and change the timing and magnitude of medical demand.
3. **Innovation:** research and technological change affect the availability, cost, effectiveness, and adoption of medical treatment.

Link across the three chapters

Public policy shapes prevention and treatment incentives; disease dynamics influence medical demand; innovation can change both the disease process and the available treatment technologies.

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